

ITCC103 (IT FUNDAMENTALS)
Week 1 Assignment

Module 1 - Unit 1 (Common Computing Devices)

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Questions:

1. Explain how the evolution of computing from the mainframe era to the current ubiquitous computing era has influenced the way we interact with technology in our daily lives. You can use examples from your own experience.

Answer: The evolution of computing has made technology much easier and more accessible. In the past, during the mainframe era, only large companies and organizations could use computers because they were expensive and very large. As technology improved, personal computers, laptops, smartphones, and tablets became available, allowing people to use technology anytime and anywhere. Today, in the ubiquitous computing era, technology is part of our daily lives through mobile apps, smart devices, and the internet. For example, I use my smartphone every day to communicate with family and friends, search for information, attend online classes, making everyday tasks faster and more convenient.

2. Our lesson mentions that Information Technology is "foundational and critical to success across virtually all industries and aspects of modern life". Provide specific examples of how IT has impacted an industry or aspect of life that is not explicitly mentioned in the material (e.g., transportation, education, or finance).

Answer: Information Technology is important because it helps improve many industries and areas of life. One example is agriculture. Farmers can now use drones, sensors, and mobile apps to monitor crops, check weather conditions, and manage irrigation. This helps increase crop production while reducing waste and saving time. IT also allows farmers to sell their products online and reach more customers. These improvements show how Information Technology continues to make work more efficient and improve people's quality of life.

3. Describe the key differences in purpose and design between a personal desktop computer and a server. In what scenarios would you choose one over the other, and why?

Answer: A personal desktop computer is meant for individual use. People often use it for studying, completing assignments, browsing the internet, playing games, or watching videos. In contrast, a server is built to provide services and store data for multiple users at the same time. It is more powerful and designed to run continuously without interruptions. I would pick a desktop computer if I only needed a device for school, entertainment, or personal tasks because it is more affordable and user-friendly. However, if I were running a business, hosting a website, or managing files that many people need to access, I would opt for a server. It is more reliable and can handle heavier workloads.

4. The computing cycle consists of input, processing, output, and storage. Choose a modern computing device, such as a smartphone or a gaming console, and describe how each of these four stages is represented in its operation.

Answer: A smartphone is a perfect example of the functioning of the computer cycle in everyday life. Touchscreen, keyboard, microphone, and camera are examples of input devices. They allow me to provide the smartphone with data and commands in order to achieve specific goals. The smartphone processor interprets the input data and instructions so that the device can perform the desired actions. Output devices such as screens, speakers, and vibrators enable the smartphone to communicate the results to me. Storage is where the smartphone saves applications, photos, videos, and other data, either in internal memory or in clouds. This is how my smartphone operates every time I try to send a message, search for information on the Internet, or complete my homework assignments.

5. What are the advantages of a desktop computer's upgradability and customization compared to a laptop? How might a user's specific needs for gaming, video editing, or portability influence their choice between these two device types?

Answer: Having a Desktop rather than a laptop is better because if some parts stop working or reached its aging state, you will be able to replace and upgrade them thus making your desktop increase its lifespan longer. While laptops are good for portability, it does have a fixed lifespan because it doesn't have a way to upgrade it further. If the user really wants a heavy workload, they should pick a desktop for raw power, while laptops have a disadvantage for power, it is great for office job related.

6. Explain the concept of the Internet of Things (IoT). How is IoT different from a traditional computer network, and what are the potential implications of a society with a vast network of interconnected physical objects?

Answer: The IoT is about making life of every people convenient, it is an automation of work to reduce strains for us. This includes like industrial machines, security systems and also wearables like smart watches for our health monitoring. Traditional network often needs a human assistant to work, while IoT's checks and automates everything, resulting in efficient daily task. If everything becomes connected, life would be much easier. but, if electricity or maybe an object is down, it would intervene the chained task that is assigned to that object, thus affecting all connected IoT objects.

7. Our lesson distinguishes between an Operating System (OS) and Applications. Based on this, explain the role of a user interface (UI) and provide examples of how a GUI differs from a CLI. How does the UI impact a user's experience with a device?

Answer: The role of UI is to make navigation easier and understandable for a user. GUI is a graphical version of CLI where it has a objects in the screen such as buttons, textfields, dropdowns and etc.. it is also dynamic and has different implementation of designs. CLI focused on non-graphical UI and instead uses texts to represent a UI for the user to interact with like entering a number to proceed or words. A well designed UI makes user able to navigate easily through the application or software, improves usability, efficiency and satisfactory for the user.

8. Our lesson discusses different types of laptops, such as 2-in-1s and Chromebooks. As a student, which of these laptop types would you choose for your studies and why? What are the potential trade-offs of your choice?

Answer: From my view, I choose 2-in-1 Laptop since it can handle heavy tasks like researching and coding. IDE, virtualization tools, AI tools and network/server simulations runs smoothly too, and you don't need to install/setup and configure a desktop OS because it already have one. The trade-off of the laptop is it has higher cost, short battery life, heavy to carry around and handling more maintenance.

9. Describe the difference between an on-premise server and a cloud server. What are the advantages and disadvantages of each from a business perspective, considering factors like flexibility and scalability?

Answer: The difference of these server is On-premise server is a physical computer hardware setup using and managing it directly in a workplace/company facilities and Cloud server is a virtual computing server that is accessed using and over the internet. The advantage and the disadvantage of this 2 is On-premise server is you have full control of the data and hardware and does not need internet access, the downside is it's expensive, hard to maintain resources/datas, require IT staff for maintenance and is vulnerable to hazards and disasters. And on Cloud server, easy to handle resources and the provider maintains the server, lower cost and accessible anywhere. The drawbacks are the ongoing months or years of subscriptions, need of internet access and less control of security.

10. Our lesson states that all digital data is represented as binary data (strings of ones and zeros). In your own words, explain why this is the fundamental representation for all types of information, including words, pictures, and video. How do transistor states relate to this concept?

Answer: All the data that you see in a computer, whether it's the text, photos or videos, is generated or stored as binary (1 & 0) because computer hardware is built from transistors. These are tiny switches that turn on and off, that is the only "language" that the hardware can understand.